INCH-POUND

(MIL-C-50652/17A (AR)
09 April 1993
SUPERSEDING
MIL-C-50652/17 (MU)
15 April 1971

MILITARY SPECIFICATION SHEET

CIRCUIT CARD ASSEMBLY: 10548638

(DELAY, FINE/FINE)

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The detail requirements of this specification sheet together with the general requirements of MIL-C-50652 comprise the complete requirements for the Fine/Fine Delay Assembly used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The Fine/Fine Delay Assembly supplies the first decoding operation of the range register bits used by the RCFD decoder, the fine/fine delay of the E/L trigger pulse during the radar search mode, and the 2 microsecond Delayed To pulse used by the range counter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
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SPECIFICATIONS

MILITARY

MIL-C-50652 - Circuit Card Assemblies, General Specifications for

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548638 - Delay, Fine/Fine

SPECIAL PACKAGING INSTRUCTION

PS10548638 - Packaging of Delay, Fine/Fine

(Copies of other Government documents, drawings, and publications, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document shall takes precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 General specifications. The assembly shall meet the requirements of MIL-C-50652.

3.2.1 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548638, all associated drawings, and with all requirements specified in applicable specifications and standards.

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3.3 Environmental service conditions. As specified in MIL-C-50652.

3.4. Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the card assembly.

3.5 Performance. For the requirements of 3.5.2 through 3.5.6.2 logic input and output levels shall be as follows:

Logic 0: 0.0 ± 0.5 Vdc level.
Logic 1: + 3.5 ± 1.0 Vdc level.

3.5.1 Input power. The input power applied between terminals P1-30, 31 and 32 (+5 V) and terminals P1-24, 25 and 26 (+5 V return) shall be 5.0 ± 0.3 Vdc, at 200 milliamperes (mA), regulated, with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.2 Input signal. Input signals to the fine/fine delay assembly shall be: T1, $\overline{T1}$, T11, F1/5, $\overline{F1/5}$, F2/5, $\overline{F2/5}$, and $\overline{F4/5}$ as shown in Figure 1. A positive pulse shall be a change in direct current (dc) level from 0.0 ± 0.5 volt (V) to 3.5 ± 1.0 V and return to 0 V. A negative pulse shall be a change in direct current (dc) level from + 3.5 ± 1.0 V to 0.0 ± 0.5 volt (V) and return to +3.5 V. For the requirements of 3.5.2 through 3.5.5 signal T1, as shown in Figure 1, and a logic 1 shall be applied to P1-34 and P1-19 respectively. For the requirements of 3.5.6 through 3.5.6.2, signal T11, as shown in Figure 1, and a logic 0 shall be applied to P1-19 and P1-34 respectively.

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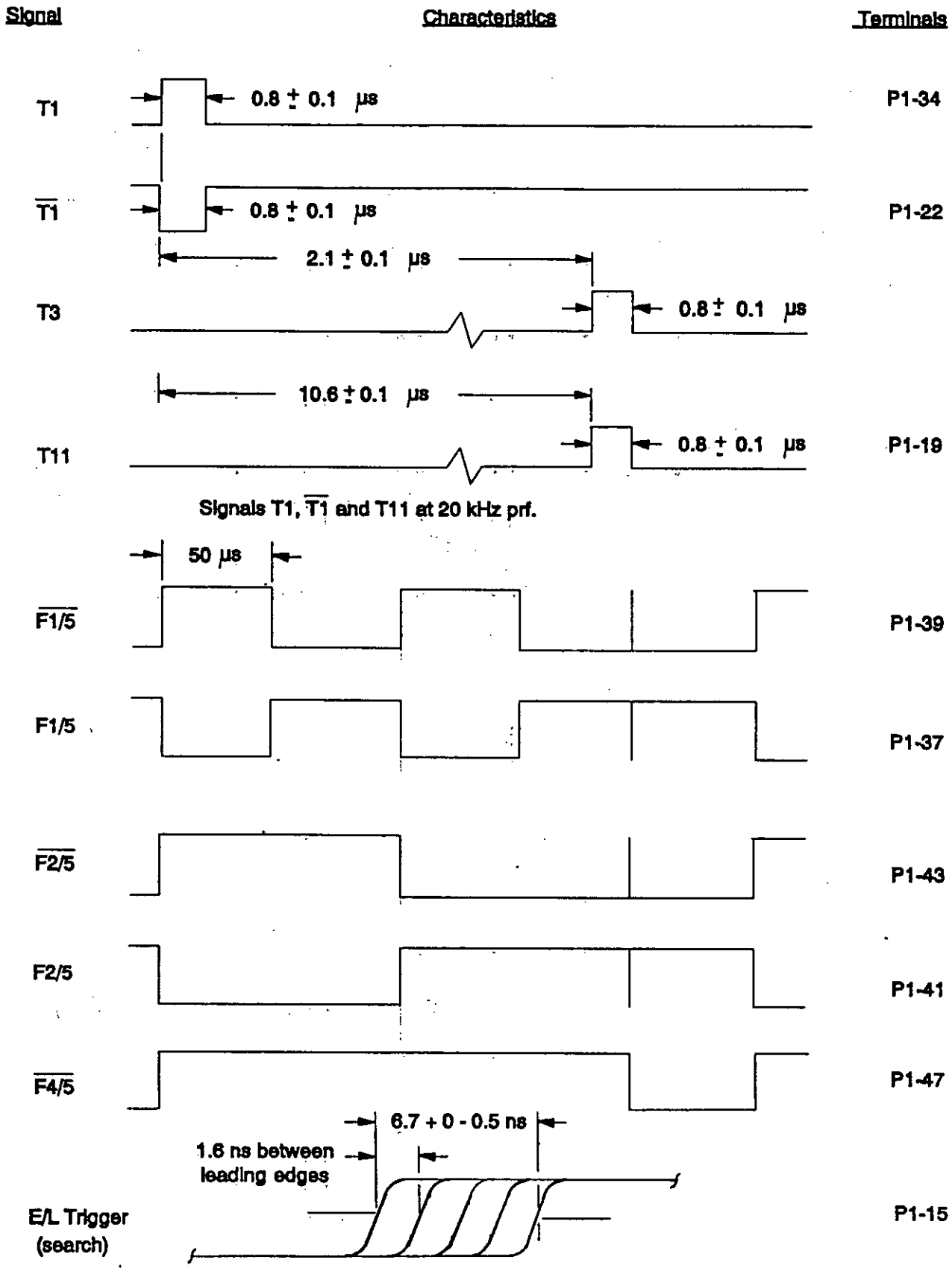


FIGURE 1. INPUT AND OUTPUT SIGNALS.

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3.5.3 RCFD decoder output. With the input levels applied sequentially, as specified in Table 1, the range control fine delay (RCFD) decoder output signals shall be as specified in Table I.

TABLE I. RCFD DECODER OUTPUT SIGNALS.

INPUT TERMINALS									SIGNAL ON OUTPUT TERMINALS							
STEP	42	51	56	44	49	46	54	52	5	6	7	8	9	10	11	12
1	0	0	0	1	1	1	1	0	0	1	1	1	0	1	1	1
2	1	0	1	1	0	1	0	0	1	0	1	1	1	0	1	1
3	0	1	0	0	1	0	1	1	1	1	0	1	1	1	0	1
4	1	1	1	0	0	0	0	1	1	1	1	0	1	1	1	0

3.5.4 To 2.0 us delay output. The delay of the positive going slope of the negative pulse at P1-36 shall be adjustable by R11 to 2.0 ± 0.1 microseconds (us) (50 percent points) from the leading edge of the input signal at P1-34. The output pulse shall be at the same 20 kilohertz (kHz) pulse repetition frequency (PRF) as the input signal.

3.5.5 To 0.2 us delay output. At P1-35 a negative pulse shall be present. It shall have an 0.2 ± 0.1 us duration and the negative going slope shall be coincident with the positive going slope of the output signal at P1-36.

3.5.6 R/L trigger output.

3.5.6.1 Fine/fine delay steps. With a logic 0 applied to P1-3, the E/L trigger output signal at P1-15 shall consist of five leading edges as shown in Figure 1. These leading edges shall be adjustable by C1 through C5 to the limits shown in Figure 1.

3.5.6.2 E/L trigger pulse parameters. The E/L trigger output signal at P1-15 shall be 0.45 ± 0.15 us wide at a 20 kHz PRF. With a logic 1 applied to P1-3 there shall be a delay of 20 to 70 nsec between the leading edge of the input signal at P1-19 and the leading edge of the E/L output signal at P1-15. With a logic 0 applied to P1-3, there shall be an additional delay of 15 to 185 nsec from the delay present with logic 1 applied to P1-3.

3.6 First Article: As specified in MIL-C-50652.

3.7 Soldering: As specified in MIL-C-50652.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification sheet.

4.2 First article. As specified in MIL-C-50652.

4.3 Inspection provisions. As specified in MIL-C-50652.

4.4 Special sampling. As specified in MIL-C-50652.

4.5 Inspection equipment. As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. As specified in MIL-C-50652.

4.5.3 Test equipment. In conjunction with 4.5 the test equipment listed in Table II shall be utilized in the performance of the applicable tests as specified in 4.6.3.

TABLE II. TEST EQUIPMENT LIST.

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ± 0.3 Vdc, regulated. Ac ripple: less than 50 mV peak-to-peak. Capacity: 200 mA.
2. Pulse generator	Three separate output pulses, 20 kHz PRF, independently variable in delay to 10.6 us ± 0.1 us, duration 0.8 ± 0.1 us, with amplitudes of +3.5 ± 1.0 Vdc into 50-ohm load.
3. Oscilloscope	Dual channel; rise time of scope plus measuring probe 5 ns or less. Sweep period 5 ns/cm or less. Input resistance: 1 megohms or greater.

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TABLE II. TEST EQUIPMENT LIST Continued...

ITEM NUMBER	DESCRIPTION
4. Sampling oscilloscope	Rise time of scope plus measuring probe 100 ps or less, sweep period 1 ns/cm or less; input resistance greater than 500 ohms, input capacitance less than 10 pF.
5. Test fixture	As specified in Figure 2; see Table III for test fixture parts list.

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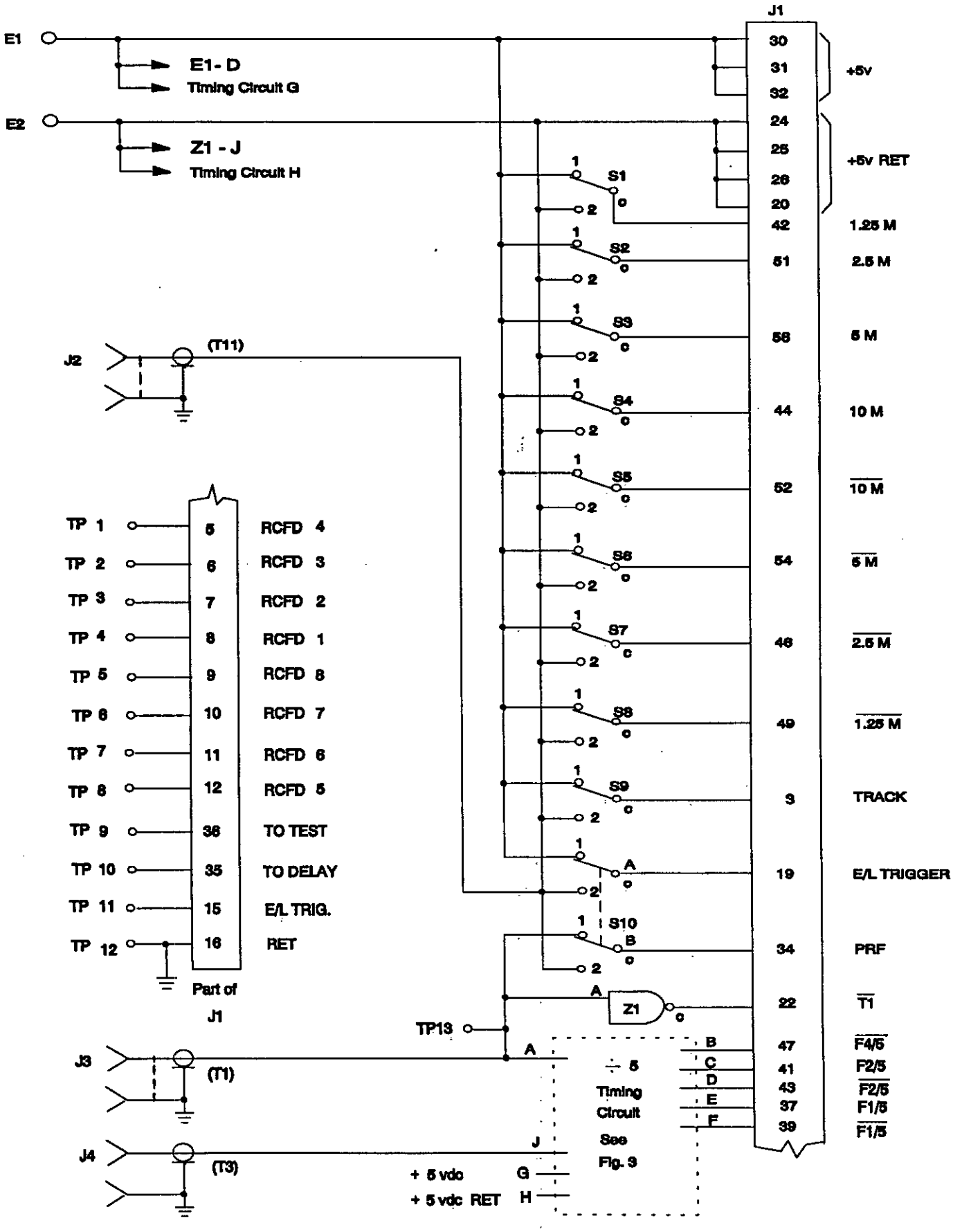


FIGURE 2. TEST FIXTURE SETUP.

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TABLE III. TEST FIXTURE PARTS LIST.

DESIGNATION	NOMENCLATURE AND DESCRIPTION
E1	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent. (1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois. (2) 500946-01, National Connector Corp. Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, PA
J2 thru J4	Connector, receptacle, UG-25B/U or equivalent.
S1 thru 9	Switch SPDT.
S10	Switch DPDT.
TPI thru TP12	Tip jack, H. H. Smith 1501 or equivalent.
Z1	Integrated circuit, SN5400-Quad 2-Input Positive-NAND Gate or equivalent.
Z2 and Z3	Integrated circuit, SN5476-Dual J-K Flip-Flop with Preset and Clear or equivalent.

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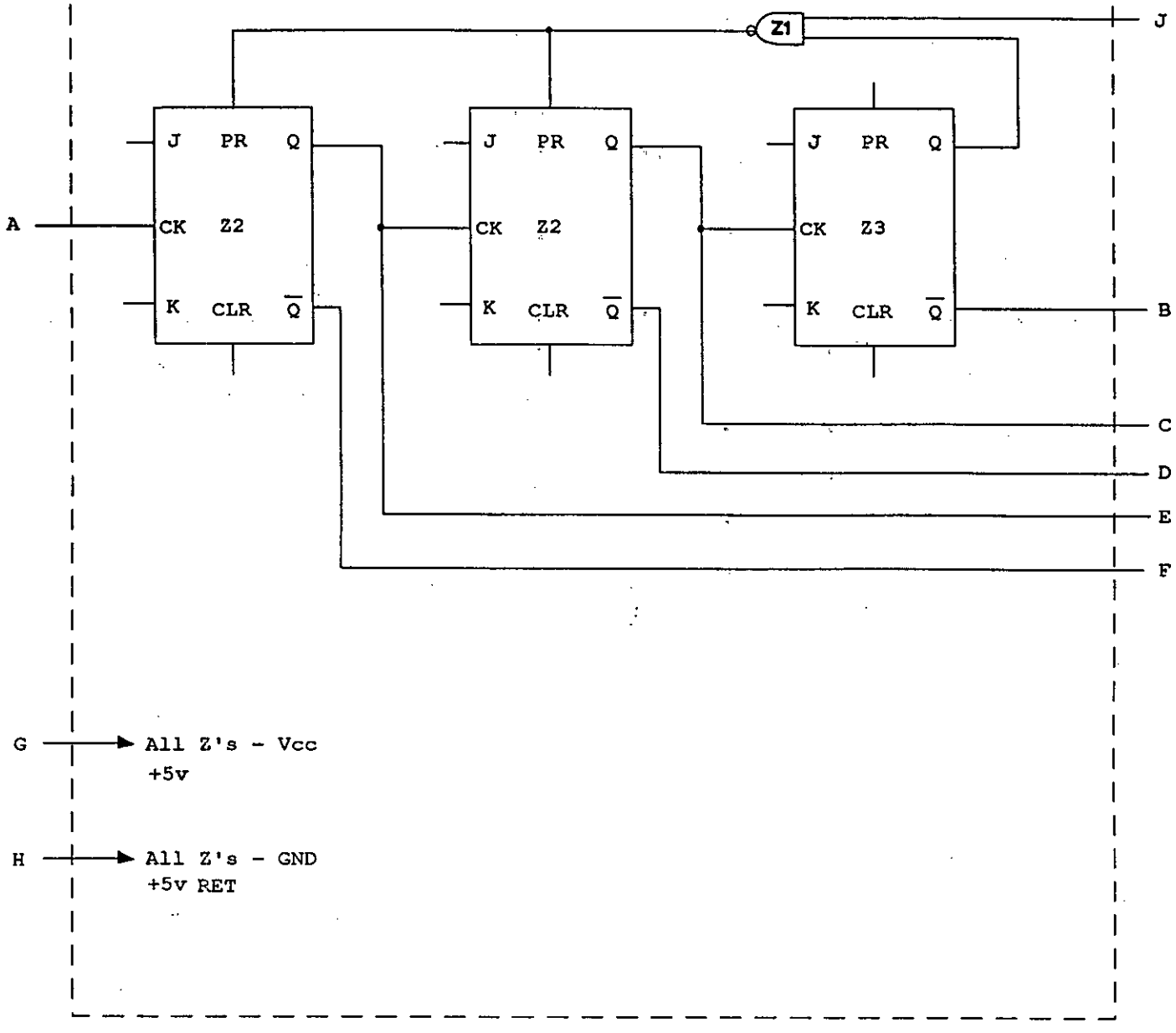


FIGURE 3. ÷ 5 TIMING CIRCUIT.

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4.6 Test methods and procedures.

4.6.1 Environmental. As specified in MIL-C-50652.

4.6.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. The selected fine/fine delay assembly shall be tested as follows: Set up equipment as shown in Figure 4. Use oscilloscope, item 3, for observation of all input and output signals as specified in 3.5.2 through 3.5.6.2 except for 3.5.6.1. For observation of output signals specified in 3.5.6.1 use oscilloscope, item 4. When specified, a logic 0 shall be observed as a 0.0 $\pm$ 0.5 Vdc level and a logic 1 as a +3.5 $\pm$ 1.0 Vdc level. Switches $\overline{S1}$ through S9 on the test fixture, item $\overline{5}$, are used to supply the logic level input signals specified in 3.5.3 through 3.5.6.2. These switches supply a logic 1 in position 1 and a logic 0 in position 2. Switch S10 in position 1 is used to apply a logic 1 level input to P1-19 and signal T1 to P1-34 for 3.5.3 through 3.5.5. In position 2, S10 applied a logic 0 level to P1-34 and signal T11 to P1-19 for 3.5.6 through 3.5.6.2.

4.6.3.1 Input power. Using item 1, apply input power as specified 3.5.1.

4.6.3.2 Input signal. Using item 2, set up and apply input signals T11, $\overline{T1}$ and T3 as shown in Figure 1 to J2, J3 and J4 respectively on test fixture, item 5. Verify that output signals from test fixture timing circuit for input signals $\overline{T1}$, F1/5, $\overline{F1/5}$, F2/5, $\overline{F2/5}$ and $\overline{F4/5}$ are as shown in Figure 1.

4.6.3.3 RCFD decoder output. Using switches S1 through S8, apply logic input levels as specified in Table I. At TP1 through TP8 observe that output signals for terminals P1-5 through P1-12 are as specified in Table I.

4.6.3.4 To 2.0 us delay output. Use oscilloscope channel A to observe input signal at TP13 (signal T1 at P1-34) and channel B to observe output signal To delayed at TP9 (P1-36). Adjust R11 on the fine/fine delay assembly to obtain an output pulse as specified in 3.5.4.

4.6.3.5 To 0.2 us delay output. Using channels A and B of the oscilloscope observe outputs at TP10 and TP9 are as specified in 3.5.5.

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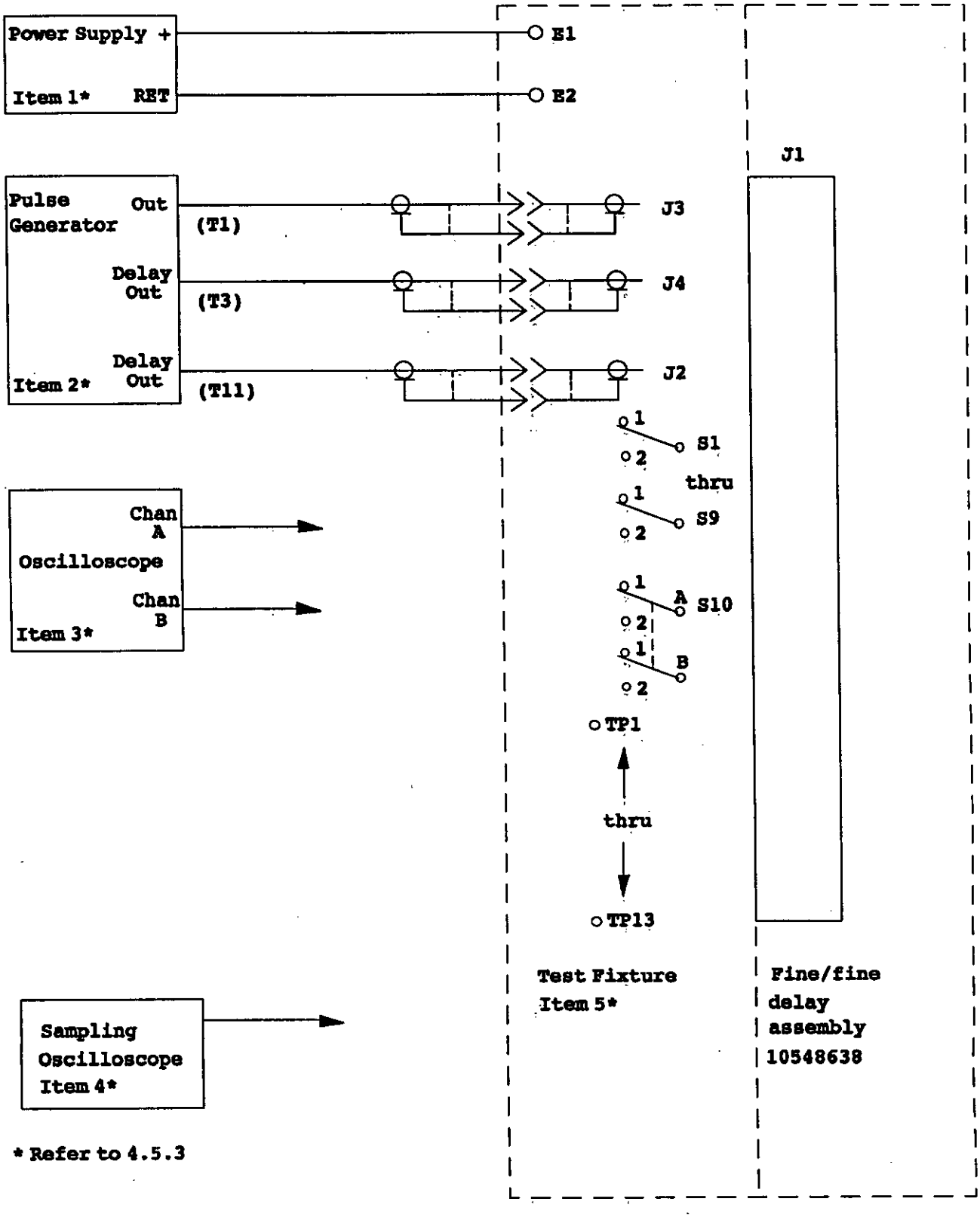


FIGURE 4. TEST SETUP.

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4.6.3.6 E/L trigger output.

4.6.3.6.1 Fine/fine delay steps. With switch S9 set to position 2, use oscilloscope, item 4, to observe the E/L trigger output signal at TP11 for compliance with Figure 1. Adjust capacitors C1 through C5 on the fine/fine delay assembly to position the five leading edges, from left to right respectively, to the limits shown in Figure 1.

4.6.3.6.2 E/L trigger pulse parameters. Use oscilloscope, item 3, to observe the E/L trigger output signal, at TP-11 is as specified in 3.5.6.2. Use switch S9 in test fixture to apply logic levels to P1-3 and observe delays of output signal as specified in 3.5.6.2.

4.7 Soldering: As specified in MIL-C-50652.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet PS10548638. The level of protection shall be as specified in the procurement document.

6. NOTES. As specified in para 6.0 of MIL-C-50652.

6.1 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-50652/17A	09 April 1993

3. DOCUMENT TITLE: CIRCUIT CARD ASSEMBLY; 10548638 (DELAY, FINE/FINE)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

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8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
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c. ADDRESS (Include Zip Code) U.S. Army ARDEC (SMCAR-BAC-S) Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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